



Teheran University of Medical Sciences

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Final Exam
International Campus, MD Program
Respiratory system
February 2023 semester

Name and Surname:

Student ID Number:

Course Coordinator: Dr. pasbakhsh

Date of Exam: July 2, 2023

Duration of Exam: 40 Minutes

MCQ Question

40

Directions:

- Each of the following questions contains one correct answer only.
- Please choose the correct answer for the MCQs in the answer sheet with pencil.



	physiology	Question time (seconds):	
	→		
1	A decrease in the oxygen pressure in the alveoli less than mmHg causes hypoxia?		
	1 80		
	2 73		
	3 60		
	4 30		

	physiology	Question time (seconds):	
	→		
2	Which of the following <u>does NOT</u> happen during inspiration?		
	1 The ribs move upward.		
	2 The diaphragm lifts up.		
	3 The anterior-posterior dimension of the chest is increased.		
	4 The vertical dimension of the thorax is increased.		

	physiology	Question time (seconds):	
	→		
3	Given your knowledge regarding the response of the pulmonary vasculature to oxygen content, where is the highest amount of blood flow through a normally oxygenated lung in a normal weight adult male that is standing upright?		
	1 To the bottom of the lung		
	2 To the top of the lung		
	3 To the midzone of the lung		
	4 There is no gradient across the lung		

	physiology	Question time (seconds):	
	→		
4	Which of the following products of the lung is responsible for increased bronchoconstriction and possible the mediators behind an acute asthma attack?		
	1 Histamine		
	2 Surfactant		
	3 Epinephrine		
	4 Vasoactive intestine peptide (VIP)		

	physiology	Question time (seconds):	
	→		
5	What is the Recoil pressure?		
	1 Plural pressure minus alveolar pressure		
	2 Alveolar pressure minus atmosphere pressure		
	3 Alveolar pressure minus plural pressure		
	4 Atmosphere pressure minus plural pressure		

	physiology	Question time (seconds):
6	The terminal bronchioles are examples of...?	→
1	Anatomical shunt	
2	Alveolar dead space	
3	Anatomical dead space	
4	Respiratory zone of airways	

	physiology	Question time (seconds):
7	Which of the following concerning average lung capacities at rest is TRUE?	→
1	TLC>VC>IC>FRC	
2	TLC>FRC>VC>IC	
3	TLC>IC>FRC>VC	
4	TLC>FRC>IC>VC	

	physiology	Question time (seconds):
8	If a patient had a progressive lung disease that required an ever increasing pressure to fill the same volume of lung, how would the lung's compliance be affected?	→
1	It would increase it.	
2	It would stay the same.	
3	It would decrease it.	
4	These variables do not affect lung compliance.	

	physiology	Question time (seconds):
9	If you blocked the blood supply to an alveolus, which of the following <u>would NOT</u> occur as a result?	→
1	The ventilation- perfusion ratio would be zero.	
2	The alveolar O ₂ pressure (PAO ₂) would be greater than normal.	
3	The alveolar CO ₂ pressure (PACO ₂) would be more than normal.	
4	The alveolar ventilation rate would be more than normal.	

	physiology	Question time (seconds):
10	Which of the following is <u>FALSE</u> concerning the relationships of the variables in diffusion of O ₂ across a membrane?	→
1	Doubling the thickness of the respiratory membrane reduces O ₂ diffusion to half.	
2	The lower the diffusion coefficient, the higher the total diffusion	
3	Increasing the alveolar concentration of oxygen would increase O ₂ diffusion	
4	Doubling the area surface of the membrane would double O ₂ diffusion	



	physiology	Question time (seconds):	
	→		
11	Which of the following is <u>NOT</u> a form by which CO ₂ can be transported in the blood?		
	1	As bicarbonate	
	2	Dissolved in the blood.	
	3	Bound to the amino end groups in hemoglobin	
	4	As carboxyhemoglobin	

	physiology	Question time (seconds):	
	→		
12	If a patient's blood carries 10 grams of hemoglobin per deciliter, what is the O ₂ carrying capacity of his blood (milliliters per deciliter)?		
	1	20.1	
	2	19.8	
	3	14.4	
	4	13.4	

	physiology	Question time (seconds):	
	→		
13	Which gas is used for estimating of "diffusing capacity of O ₂ " in the lung?		
	1	CO ₂	
	2	CO	
	3	O ₂	
	4	He	

	physiology	Question time (seconds):	
	→		
14	What is the cause of "physiological shunt"?		
	1	A decrease in ventilation/perfusion ratio (VA/Q) to zero	
	2	Admixture of pulmonary vein blood with bronchial vein blood	
	3	An increase in VA/Q ratio 2.5 times more than normal amount	
	4	A decrease in the cardiac output to half	

	physiology	Question time (seconds):	
	→		
15	The affinity of hemoglobin for O ₂ increases by?		
	1	A decrease in pH	
	2	An increase in CO ₂ pressure	
	3	A decrease in DPG concentration	
	4	An increase in body temperature	

	physiology	Question time (seconds):
16	The duration of exhalation:	→
1	Increases by strong activation of pneumotaxic center	
2	Is less than duration of inhalation during normal and quiet respiration	
3	Is affected by ventral respiratory group (VRG) during normal respiration	
4	Decreases when respiratory rate increases	

	physiology	Question time (seconds):
17	Which one of the following neurotransmitters is <u>NOT</u> released from glomus cell in carotid bodies?	→
1	Dopamine	
2	Acetyl choline	
3	ATP	
4	VIP	

	physiology	Question time (seconds):
18	What changes are happening in the breathing pattern by the destruction of the pneumotaxic center?	→
1	The inspiration period becomes prolonged and deep	
2	The breathing rate increases	
3	The tidal volume decreases	
4	It doesn't make a change	

	physiology	Question time (seconds):
19	How much carbon dioxide is expelled by the lungs every minute in rest state?	→
1	4 ml	
2	21 ml	
3	200 ml	
4	250 ml	

	physiology	Question time (seconds):
20	Why is the oxygen pressure of the systemic arterial blood (95 mmHg) lower than the oxygen pressure of the alveolar air (104 mmHg) under normal conditions?	→
1	The presence of physiological dead space in the apex of the lung	
2	Venous admixture in the pulmonary circulation system	
3	The ratio of ventilation to blood flow (V/Q) is less than one at the base of the lung (V/Q=0.6)	
4	Less compliance of systemic arteries compared to pulmonary arteries	



	physiology	Question time (seconds):
	→	
21	What is the "diffusion coefficient" of the gas?	
	1 Gas solubility	
	2 The square root of the molecular weight of the gas	
	3 The ratio of solubility to the square root of the molecular weight of the gas	
	4 The ratio of solubility to the distance of diffusion	

	physiology	Question time (seconds):
	→	
22	Which one of the following factors decreases in the respiratory system during exercise?	
	1 Diffusing capacity of oxygen	
	2 The amount of blood flow in the top of the lungs	
	3 The alveolar ventilation rate	
	4 The duration of blood passage through the pulmonary capillaries	

	Anatomy	Question time (seconds):
	→	
23	All of the following bones are seen in lateral wall of nasal cavity except...?	
	1 Lacrimal	
	2 Maxilla	
	3 Vomer	
	4 Palatin	

	Anatomy	Question time (seconds):
	→	
24	Which of the following bones is seen in medial border of posterior opening of nasal cavity ?	
	1 Vomer	
	2 Maxilla	
	3 Palatin	
	4 Sphenoid	

	Anatomy	Question time (seconds):
	→	
25	Which of the following artery is seen in incisive canal ?	
	1 Sphenopalatin	
	2 Greater palatin	
	3 Anterior ethmoidal	
	4 Superior alveolar	

	Anatomy	Question time (seconds):
26	Which of the following bones participate in both roof and medial surface of nasal cavity?	→
	1 Maxilla	
	2 Sphenoid	
	3 Palatin	
	4 Lacrimal	

	Anatomy	Question time (seconds):
27	Which of the following paranasal sinus open in superior meatus?	→
	1 Anterior group of the ethmoidal cell	
	2 Posterior group of the ethmoidal cell	
	3 Maxillary sinus	
	4 Frontal sinus	

	Anatomy	Question time (seconds):
28	Which of the following options related to larynx is correct ?	→
	1 The lower chamber of the laryngeal cavity is called the vestibule.	
	2 The vocal ligament is on the upper and free side of the quadrat membrane.	
	3 The vocal fold is located at the top of the arytenoid cartilage.	
	4 True vocal folds are located at the bottom of the middle chamber of the larynx cavity.	

	Anatomy	Question time (seconds):
29	All of the following options are correct except....?	→
	1 Arytenoid cartilages go to abduction position during breathing.	
	2 Arytenoid cartilages go to adduction position at the time of speaking.	
	3 Vocal cartilages are near the midline at the time of swallowing.	
	4 Arytenoid cartilages go to adduction position in deep breathing.	

	Anatomy	Question time (seconds):
30	All of the following options are correct except....	→
	1 The internal branch of the superior laryngeal nerve is responsible for the upper part of the laryngeal cavity.	
	2 The transverse arytenoid muscle is the only unpaired intrinsic muscle of the larynx	
	3 Laryngeal recurrent nerves innervate the transvers pyramidal muscles in the larynx	
	4 The external branch of the superior laryngeal nerve passes through the hole of the thyrohyoid membrane	



	Anatomy	Question time (seconds): →
31	Which of the following muscles is a part of thyroarytenoid muscle? Vecalis muscle	
	1 Cricothyroid	
	2 Vocalis muscle	
	3 oblique arytenoid	
	4 Transverse arytenoid	

	Anatomy	Question time (seconds): →
32	Uncinate process is seen in which of the following bones?	
	1 Maxilla	
	2 Vomer	
	3 Ethmoid	
	4 Sphenoid	

	Anatomy	Question time (seconds): →
33	Nasopalatin nerve is branch of which of the following nerve?	
	1 Maxillary	
	2 Anterior Ethmoidal nerve	
	3 Greater palatin nerve	
	4 Posterior inferior nasal	

	embryology	Question time (seconds): →
34	All the following areas originate from meta nephrosis tissue except	
	1 Proximal Convoluted tubule	
	2 Nephron	
	3 Distal Convoluted tubule	
	4 minor calyces	

	embryology	Question time (seconds): →
35	Which of the following option is correct?	
	1 The proximal end of the urorectal septum forms the perineal body	
	2 Pronephrosis appears in the fifth week	
	3 The middle part of the urogenita sinus forms the bladder	
	4 Meta-nephrosis makes permanent kidney	

Embryology

متن سوال (لطفا در باکس زیر تایپ شود)		
.In which period of lungs development alveolar duct appear		34
Alveolar phase	الف	
Pseudoglandular phase	ب	
Canalicular phase	ج	
Saccular phase	د	
		منبع:

متن سوال (لطفا در باکس زیر تایپ شود)		
In which time of development the second epithelial alveolar cells appear?		35
3th month	الف	
4 th month	ب	
5 th month	ج	
6 th month	د	
		منبع:



	Histology	Question time (seconds):
36	In which of the following is observed the Histology Structure Of Vocal Cord?	→
	1 Vestibuli Zone	
	2 Olfactory Zone	
	3 Pharynx	
	4 Respiratory Zone	

	Histology	Question time (seconds):
37	Which part of the lung tissue has a simple squamous cells?	→
	1 Bronchus	
	2 Terminal Bronchiole	
	3 Alveolar	
	4 Bronchiole	

	Histology	Question time (seconds):
38	What is the name Macrophage Alveolar Of Lung?	→
	1 Pneumocyte I	
	2 Pneumocyte II	
	3 Pneumocyte III	
	4 Pneumocyte IV	

	Histology	Question time (seconds):
39	Where is Mucosa & Serosa Gland In Around of Trachea ?	→
	1 Adventitia	
	2 Mucous	
	3 Muscularis	
	4 Submucosa	

	Histology	Question time (seconds):
40	In which of the following is observed the Unicell Gland Or Goblet Cell?	→
	1 Vestibuli Zone	
	2 Respiratory Zone	
	3 Terminal Bronchiole	
	4 Alveolar Duct	